# George Zhang

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Canadian & Hong Kong citizen · eligible for U.S. internships (J-1); TN-eligible post-graduation

## Education

**University of Waterloo** — Waterloo, ON

*Bachelor of Computer Science (Honours, Co-op) — Artificial Intelligence Specialization* — *Sep 2024 – May 2029*

- Relevant coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Optimization, Logic & Computation, OOP (C++).

## Experience

**Ford Motor Company** — Waterloo, ON

*Software Developer Intern — IVI Phone Apps* — *May 2026 – Aug 2026*

- Coordinated cross-team development of an in-progress MAU PowerBI dashboard for phone apps, aligning requirements and data sources between the IVI team and the analytics team.
- Spearheaded two automated workflows for the team's LLM-based log-analysis playbook — tag-differential-and-chunking updates, and a sync job publishing the playbook directly to Confluence.
- Shipped features across three in-vehicle apps (Dialer, Settings, Messenger) on Android Automotive OS, Ford's production infotainment platform; closed all 64 completed issues, a 100% completion rate on finalized work.

**University of British Columbia** — Vancouver, BC

*Undergraduate Research Assistant — Centre for Brain Health* — *Jun 2025 – Aug 2025*

- Designed activity-heatmap dashboards visualizing patient exercise data that cut clinician monitoring overhead by 30%; presented at the 11th Singapore International Parkinson Disease & Movement Disorders Symposium.
- Led development of two full-stack hand-rehabilitation apps within PIKA, an AI Parkinson's-care platform deployed at Vancouver Coastal Health.
- Built a real-time computer-vision pipeline (Python, OpenCV, MediaPipe) detecting 7 hand-exercise gestures at ~90% accuracy with under 100 ms latency, fully on-device to meet PIPEDA/GDPR requirements.

## Projects

**Code-Aware RAG System** | Python, Pandas, embeddings — github.com/TheYellowDuck/RAG-codebase

- Designed the evaluation framework for a retrieval-augmented-generation system: a metrics suite (recall@k, MRR, NDCG) with bootstrap confidence intervals and paired significance tests, treating every design choice as a measured ablation.
- Analysis drove decisions: the LLM reranker significantly improved ranking (MRR/NDCG, $p < 0.001$) while a cross-encoder proved net-negative on code and was cut — retrieval fused dense + BM25 (RRF) with a code graph and personalized PageRank; provider-agnostic (Anthropic/OpenAI/local).

**Autonomous LLM Web Agent** | Python, Playwright — github.com/TheYellowDuck/web-agent

- Designed a ReAct + reflection agent (browser accessibility tree, provider-agnostic across Claude/GPT/Gemini) with a WebArena / Online-Mind2Web eval harness; showed the same agent scores 0.43 vs 0.61 by scorer alone, isolating measurement from real capability.

**Interactive Portfolio & Content Pipeline** | TypeScript, Next.js, Redis — georgezhang.ca

- Built a personal site on a custom HTML5 Canvas engine with an automated daily data pipeline (GitHub Actions) that fetches, transforms, and publishes fresh content, plus multiplayer "ghost trails" persisted in Upstash Redis.

## Technical Skills

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| **Data & Analysis** | SQL, Pandas, NumPy, scikit-learn, Matplotlib, statistics, significance testing, bootstrap CIs, data visualization |
| **Languages** | Python, C++, Java, Kotlin, TypeScript/JavaScript |
| **ML / AI** | PyTorch, Transformers, RAG, LLM agents, computer vision (OpenCV, MediaPipe), embeddings, LLM-as-judge eval |
| **Tools** | Git, Linux, Docker, GitHub Actions, Playwright, Anthropic/OpenAI, Power BI, Confluence |
| **Concepts** | Statistical Evaluation & Experiment Design, Data Structures & Algorithms, Dashboarding, OOP |

## Awards & Competitive Programming

- President's Scholarship of Distinction & Entrance Scholarship in Mathematics, University of Waterloo
- ICPC Local (Waterloo) — Rank 28 / ~100 · 470+ LeetCode & 220+ DMOJ solved (1,497 points)
- Canadian Computing Competition — Junior: perfect score, Senior: top quartile · AIME Qualifier · COMC Distinction
- RoboCup International Junior Maze Rescue 2023 (Captain) — 2nd Place Supergroup, 17th Individual